Guidelines for Obtaining IRB approval for Human Subject Imaging Research Using Verasonics Ultrasound Systems

APPLICATION NOTE | D00055

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Summary

This document provides recommendations to obtain IRB approval to use the Vantage™ Research Ultrasound System with human subjects. These recommendations are based on insights from prior successful (and unsuccessful) applications.

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1. Introduction

The guidelines summarized in this document are a set of recommendations for researchers based on information provided by research groups in the US that have obtained Investigational Review Board (IRB) approval for human subject imaging research using Verasonics systems. Note that depending on your local institutional policies, these guidelines may be either insufficient or overly restrictive. This document has not been submitted to any regulatory body for comment or approval.

The basic approach to applying for human studies approval is to provide a letter to the institution's local IRB panel making a case that the Vantage System is comparable to a conventional ultrasound imaging system in terms of patient exposure and should be assessed for safety in that context.

Early attempts at obtaining Investigational Device Exemption (IDE) approval were not successful as the emphasis was placed on explaining the flexibility of the system as a research tool, which raised more concern than it alleviated, particularly when new imaging modes were discussed. For example, the use of unfocussed beams, which ensonify wide areas, are unconventional and can be described in ways that are either alarming or reassuring.

Ultimately, the researcher and the IRB are responsible for the safety of human subjects. Quantitative acoustic field measurements are the rational basis for safety assessment and are strongly recommended as part of an application.

2. Guidelines

In 2019, Verasonics completed calibrated field measurements for the “High-Quality Image” subset of example scripts provided with every Vantage V4.2+ software installation, for specific transducers of type P4-2v, C5-2v, L11-5v, and L12-3v. These measurements were analyzed to provide tabulated values of Mechanical Index (MI) vs. programmed drive voltage, in accordance with AIUM NEMA UD2 (2007). The corresponding tables and plots of the measured results are provided in the *High Image Quality* folders for those specific transducer models, located in the *Example_Scripts/Biomedical/Vantage 128* and *256/UTA-260-S and 260-D* folders.

The above data are a useful reference to include in your application, however, it is applicable specifically to the ultrasound beam for which it was measured. Consequently, you should include the following disclaimer with each set of data.

> *The MI data provided below is for reference only. It is the user's responsibility to determine whether to perform additional measurements depending on how the information is used.*
>
> *The measurements were taken using one transducer on one Vantage System. Verasonics did not evaluate the variability in the measurements between different transducers nor between different Vantage Systems, so the MI values may differ for measurements conducted with a different transducer and/or Vantage System. Moreover, measurement errors may be introduced when using a different hydrophone or measurement system.*

In addition to the usual non-instrumentation components of an IRB approval for human studies—such as descriptions of the clinical exam and the patient consent form—successful applications have included the following information:

- A general statement indicating that the Vantage ultrasound data acquisition system, combined with a researcher-provided commercial transducer, and the researcher's software (acquisition sequence control program, that is, so-called "SetUp scripts"), produces ultrasound signals and exposures that are similar to those produced by approved clinical systems (i.e., it produces ultrasound exposures that are similar to those produced from an "ordinary" ultrasound system, and under FDA limits).
- A description of the ultrasound exposure regime and the imaging sequence as programmed in the script(s). The selected field beam patterns (simulations) and calibrated hydrophone field measurements (associated with Thermal Index (TI) and MI calculations) indicate that the exposures generated by the system are within FDA limits. Users can perform their own simulations; however, the **`showTXPD.m`** transmit beam intensity function included in the system software should be sufficient to identify the most intense field regions of interest. For applications such as elastographic radiation force "push" sequences that include extended transmissions using TPC Profile 5, water tank acoustic power and intensity (AP&I) measurements are required.
- A display of the TI and MI on screen via custom graphical annotations, unless the exposure levels are sufficiently below the FDA limits, as is often the case for any broad beam transmit fields.
- A list of steps to ensure that clinical users can only use the software if it is evaluated and tested, and that clinical users cannot configure settings that exceed the established operating limits. While this is primarily a software configuration management process, it also indicates that the FDA limits have been assessed and that the clinical researcher is unable to exceed them by adjusting settings on the GUI. For example, the MATLAB® Compiler can be used to compile a software application into an executable with encrypted source code—this compiled code cannot be inspected or modified. Verasonics has documentation and examples describing how to compile user code using the MATLAB Compiler. Please contact Verasonics support (support@verasonics.com) if you are interested in receiving this example.
- An electrical inspection/ certification that must be conducted at your institution with your system. This is a test of electrical leakage current and ground quality/integrity for the entire system, including your chosen PC and transducer. Because the construction of the system uses medical power supplies and medical grade EMI shielding, these tests have always passed without the use of an isolation transformer, even with the computer connected to the Vantage via the PCIe cable and plugged into typically grounded wall power receptacles. Researchers need to inquire with their local instrumentation safety certification testing facility to arrange for testing at their site.

3. Risk Analysis and Built-in Utilities

The *Vantage Risk Analysis* document also provided in the *Safety* folder follows the FDA guidelines for IDE applications and describes hazards, their severity, and mitigations that Verasonics has undertaken. The document supports the idea that Verasonics has taken steps to mitigate all significant risks associated with electrical safety. The Risk Analysis document addresses extended transmit regimes that are possible with the Extended Transmit System option, including the external power supply HIFU Option, which is capable of ablative energies. The *Vantage Risk Analysis* clearly identifies issues that remain the user's responsibility, since the system can be programmed by the user to do a wide range of applications beyond Verasonics' control, and that can create potentially hazardous exposures when user configuration settings exceed FDA safety limits. The applicant may benefit from addressing some of the potential hazards they have control over in a similar Risk Analysis document of their own, depending on local IRB requirements.

In addition, to protect the Vantage hardware in high power modes, Verasonics has developed software that evaluates various hardware operating state parameters, as programmed by the user's sequence, and tests those operating parameters against predefined limits. The software is available to the user for enforcement of AP&I limits (of their choosing) as well (see the **`Utilities/TXEventCheck.m`** function). This approach can supplement normal programmable limits on the HV supply (transducer transmit voltage) and can make use of the operational safeguards built into the system for any chosen protocol. Verasonics may need to provide some help if this approach is used; however, there is relevant documentation for the **`TXEventCheck`** function in the Sequence Programming Manual and using the help function.

4. Conclusion

These guidelines have evolved over time; they were originally much more ambitious than needed and may still be too conservative for many applications. Verasonics does not have copies of the IRB applications submitted by various groups. In our experience and from user reports, each IRB application is different. Meeting in advance with the human subjects committee representative was helpful to the groups with whom we have had the most experience. When the IRB panel has no ultrasound expertise, either having an independent expert sit in or write a letter of support is recommended.

Note: The Vantage Risk Analysis document is included in the documentation package provided in the Vantage software installation folder: Documentation/Safety.

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